

Date: \_\_\_\_\_

Day: \_\_\_\_\_

## ANOVA Test:

used to jointly analyze multiple populations.

Suppose we have  $K$ -independent population / treatment

\* Normally distributed

\* same variance ( $\sigma^2$ )

| Treatment   | 1              | 2              | ... | K              |
|-------------|----------------|----------------|-----|----------------|
|             | $y_{11}$       | $y_{12}$       | ... | $y_{1K}$       |
|             | $y_{21}$       | $y_{22}$       | ... | $y_{2K}$       |
|             | $\vdots$       | $\vdots$       | ... | $\vdots$       |
|             | $y_{N,1}$      | $y_{N,2}$      |     | $y_{N,K}$      |
| Sample mean | $\bar{y}_{1.}$ | $\bar{y}_{2.}$ |     | $\bar{y}_{K.}$ |
| Final Mean  | $\bar{y}_{..}$ |                |     |                |

$y_{11}$  : 1<sup>st</sup> sample of pop 1

$K$  : Number of pop

$y_{ij}$  =  $i$ <sup>th</sup> sample of  $j$ <sup>th</sup> pop

Date: \_\_\_\_\_

Day: \_\_\_\_\_

$$y_{ij} = \mu_j + \epsilon_{ij}$$

$$y_{ij} = \frac{1}{K} \sum_{j=1}^K \mu_j + \alpha_j + \epsilon_{ij}$$

Grand Mean

Effect of  $j$ th treatment

$$\sum_{j=1}^K \alpha_j = 0$$

### Hypothesis:

both  
Acceptable

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_K$$

$H_1$  : At least two means are not ~~different~~ equal

OR

$$H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_K = 0$$

$H_1$  : At least two  $\alpha$ 's are not zero.

### Critical Region:

$$F > F_{\alpha}(V_1, V_2)$$

$$V_1 = K - 1$$

$$V_2 = N - K$$

$$\mu_j = \frac{1}{N} \sum_{i=1}^N y_{ij}$$

$K$ : Number of population

$N$ : Sum of all  $n$

$$N = \sum_{j=1}^K n_j$$

**Test statistics:**

**1) Same Sample Size:**

$$F_0 = \frac{SSA / K - 1}{SSE / K(n-1)}$$

$$SSA = n \sum_{j=1}^K (\bar{y}_{j\cdot} - \bar{y}_{\cdot\cdot})^2$$

$$SSE = \sum_{j=1}^K \sum_{i=1}^N (y_{ij} - \bar{y}_{j\cdot})^2$$

$n$  = sample size (if  $S$  populations have  
9 samples then  $n$  will be 9)

**2) Different sample size:**

$$F_0 = \frac{SSA / K - 1}{SSE / N - K}$$

$K$ : Number of population

$N$ : Sum of all  $n$

$$N = \sum_{j=1}^K n_j$$



$$SSA = \sum_{j=1}^K n (\bar{Y}_{j\cdot} - \bar{Y}_{\cdot\cdot})^2$$



different (depend on each  
pop size)

$$SSE = \sum_{j=1}^K \sum_{i=1}^N (y_{ij} - \bar{Y}_{j\cdot})^2$$

$n$  = Different for each pop.

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**EXAMPLE 13.1**

Test the hypothesis  $\mu_1 = \mu_2 = \dots = \mu_5$  at the 0.05 level of significance for the data of Table 13.1 on absorption of moisture by various types of cement aggregates

Table 13.1: Absorption of Moisture in Concrete Aggregates

| Aggregate: | 1      | 2      | 3      | 4      | 5      |        |
|------------|--------|--------|--------|--------|--------|--------|
|            | 551    | 595    | 639    | 417    | 563    |        |
|            | 457    | 580    | 615    | 449    | 631    |        |
|            | 450    | 508    | 511    | 517    | 522    |        |
|            | 731    | 583    | 573    | 438    | 613    |        |
|            | 499    | 633    | 648    | 415    | 656    |        |
|            | 632    | 517    | 677    | 555    | 679    |        |
| Total      | 3320   | 3416   | 3663   | 2791   | 3664   | 16,854 |
| Mean       | 553.33 | 569.33 | 610.50 | 465.17 | 610.67 | 561.80 |

**Example:**

**Hypothesis:**

$$H_0 = \mu_1 = \mu_2 = \mu_3 = \dots = \mu_5$$

$H_1 =$  At least two means are not ~~different~~ equal

**Test statistics:**

(same sample size  $n$ )

$$F = \frac{SSA / K - 1}{SSE / (K(n - 1))}$$

$$SSA = n \sum_{i=1}^K (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$$

$$\bar{y}_{1\cdot} = 553.3$$

$$\bar{y}_{2\cdot} = 569.33$$

$$\bar{y}_{3\cdot} = 610.5$$

$$\bar{y}_{4\cdot} = 465.16$$

$$\bar{y}_{5\cdot} = 610.6$$

$$\bar{y}_{\cdot\cdot} = 561.778$$

$$SSA = 6 (14221.3) = 85356$$

$$SSE = \sum_{j=1}^K \sum_{i=1}^n (y_{ij} - \bar{y}_{j\cdot})^2 = 124021$$

The values

**Critical Region:**

$$F_0 = \frac{85356/4}{124021/5(6-1)}$$

$$F_0 = 4.3$$

**Critical Region:**

$$F_{0.05}(4, 25) = 2.76$$

$$4.3 > 2.76$$

Reject  $H_0$

It means that at least two means are not ~~different~~ equal

**EXAMPLE 13.2:**

Part of a study conducted at Virginia Tech was designed to measure serum alkaline phosphatase activity levels (in Bessey-Lowry units) in children with seizure disorders who were receiving anticonvulsant therapy under the care of a private physician. Forty-five subjects were found for the study and categorized into four drug groups:

G-1: Control (not receiving anticonvulsants and having no history of seizure disorders)

G-2: Phenobarbital

G-3: Carbamazepine

G-4: Other anticonvulsants

From blood samples collected from each subject, the serum alkaline phosphatase activity level was determined and recorded as shown in Table 13.4. Test the hypothesis at the 0.05 level of significance that the average serum alkaline phosphatase activity level is the same for the four drug groups.

Table 13.4: Serum Alkaline Phosphatase Activity Level

| G-1    |        | G-2    | G-3    | G-4    |
|--------|--------|--------|--------|--------|
| 49.20  | 97.50  | 97.07  | 62.10  | 110.60 |
| 44.54  | 105.00 | 73.40  | 94.95  | 57.10  |
| 45.80  | 58.05  | 68.50  | 142.50 | 117.60 |
| 95.84  | 86.60  | 91.85  | 53.00  | 77.71  |
| 30.10  | 58.35  | 106.60 | 175.00 | 150.00 |
| 36.50  | 72.80  | 0.57   | 79.50  | 82.90  |
| 82.30  | 116.70 | 0.79   | 29.50  | 111.50 |
| 87.85  | 45.15  | 0.77   | 78.40  |        |
| 105.00 | 70.35  | 0.81   | 127.50 |        |
| 95.22  | 77.40  |        |        |        |



Example:

Hypothesis:

$$H_0 = \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$H_1$ : At least two of the

Test Statistics:

Different sample size so,

$$F = \frac{SSA / K - 1}{SSE / N - K}$$

$$N = 20 + 9 + 9 + 7$$

$$N = 45$$

$$K = 4$$

$$SSA = \sum_{j=1}^K n (\bar{y}_{j\cdot} - \bar{y}_{\cdot\cdot})^2$$

$$SSE = \sum_{j=1}^K \sum_{i=1}^N (y_{ij} - \bar{y}_{j\cdot})^2$$

$$N - K = 45 - 4 = 41$$

$$F = 3.57$$



Critical Region:

$$F_{0.05}(44, 41) = 2.61$$

$$3.57 > 2.61$$

Reject  $H_0$ .